

Communication Tone, Pull Request Outcomes, and Contributor Return in Open Source Software

Part 2: Mid-term Presentation

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Part 1

What question are we tackling?

Part 1

Research Question and Background

Central question

Are communication signals in PR discussion associated with **merge outcomes** and **contributor return**?

Four research questions

- RQ1 Which signals are associated with **PR merge**?
- RQ2 Which signals are associated with **author return** (30 / 90 / 180 days)?
- RQ3 Do **newcomers** experience or respond to discussion differently?
- RQ4 Does **politeness** moderate the association between harsh language and return?

Study framing

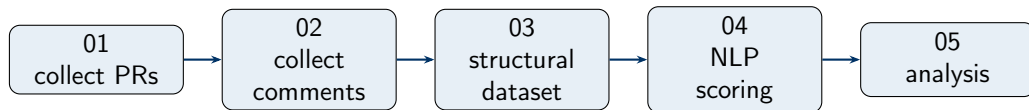
Observational: *which discussion signals associate with outcomes?*

Part 2

What research progress has been made?

Part 2

Pipeline Built — Data Collected



Dataset scale

- **12,098** focal PRs (Jan 2024 – Mar 2025)
- **3** repositories: Kubernetes, Next.js, scikit-learn
- **2,210** distinct PR authors
- **100k+** comments scored with NLP

Signals computed per PR

- toxicity (mean, max, binary flags)
- sentiment
- anger
- politeness

Status

Full pipeline complete; preliminary analysis running.

Part 2

What the Data Looks Like So Far

Descriptive picture

- Merge rate: **~76%** across repositories
- Return rates are high — majority of authors return within 90 days
- Overt toxicity is **very sparse**: <1% of PRs exceed common thresholds
- Sentiment slightly negative; anger and politeness vary more widely

Outcome rates (preliminary)

Outcome	Rate
Merge	75.7%
Return 30d	76.8%
Return 90d	81.3%
Return 180d	83.5%
Toxic ≥ 0.05	1.0%

Early pattern

Preliminary models: **contributor experience** dominates return. Communication signals carry more signal for **merge** than for return.

Part 3

Challenges encountered and adjustments made

Part 3

Operationalizing Toxicity: From Average to Maximum

What the data revealed

Mean toxicity per PR is near zero almost everywhere. Averaging washes out rare but impactful hostile comments.

Why max toxicity

A single hostile comment can affect a contributor's experience. The average never reveals it; the maximum does.

Design choice

Primary emphasis placed on:

- **max toxicity** per PR — captures worst-case exposure
- **binary indicators**: any comment ≥ 0.05 , any ≥ 0.10
- both used as robustness checks against

Broader signal picture

Sentiment, anger, and politeness vary more widely than toxicity and carry more usable signal across all RQs.

Part 3

Classifying Experience and Controlling for Confounds

Newcomer definition requires care

A first-PR rule mislabels contributors who were active before the collection window.

6-month lookback window

Collection extended to **Jul 2023**; focal PRs start **Jan 2024**.

The Jul–Dec 2023 buffer provides prior history to classify contributors reliably.

Same design ensures **complete 180-day return labels** for all focal PRs.

Known limitation: NLP domain

Pre-trained models were trained on social media. In code review, *“this is completely broken”* is technical criticism, not hostility. Scores are used directly; this is acknowledged as a limitation.

Added controls

Contributor history, discussion size, first-response delay, and repository fixed effects included to isolate the communication signal.

Part 4

Future research plans

Part 4

What Comes Next

Analysis to complete

- Finalize logistic regressions for all 4 RQs with full controls
- Run binary toxicity robustness checks
- Test newcomer $\times$ toxicity interaction (RQ3)
- Confirm politeness moderation result (RQ4)

Interpretation and write-up

- Present findings clearly for each RQ
- Keep only figures that communicate a clear result
- Discuss limitations:
 - PR-only return definition
 - Sparse toxicity limits power
 - NLP-domain mismatch
 - Observational design only

Expected findings

Communication signals associate more strongly with **merge outcome** than with contributor return. Contributor experience and history are the dominant predictors of return.

Thank you

Questions?